

Topic 3. SPACE EXPLORATION.

When I orbited the Earth in a spaceship, I saw for the first time how beautiful our planet is. Mankind, let us preserve and increase this beauty, and not destroy it! (Yuri Gagarin, Russian cosmonaut, the first man in space)



<https://www.youtube.com/watch?v=LfycThKOX8c> The First Human in Space (13 min)

Earth is the best planet in our solar system. We go to space to save Earth (Jeff Bezos)

DISCUSSION/SPEAKING PRACTICE

What are Spin-offs?

Definition:

a collateral or derived product or effect

a by-product

also: a number of such products

something that has been developed from something else

e.g. the spin-off from the space program

Do you know any spin-off devices that were originally developed for NASA purposes and that we now use on a daily basis?

You can find some here: <https://spinoff.nasa.gov>

Might be of interest: BBC Business Daily podcast, July 28, 2021

The billionaire space race <https://www.bbc.co.uk/programmes/w3ct1jnt>

WATCH THE FOLLOWING VIDEO

<https://www.youtube.com/watch?v=5OLttelwwNs> Three Men Lost in Space – The Apollo 13 Disaster (23 min 30 sec), 2021

Answer the following questions based on the video:

1. What does NASA stand for?

2. What are some of NASA inventions mentioned in the intro?

3. What makes space travel perilous?

4. When did the Apollo 13 disaster take place?

5. When did the first moon landing occur?

6. What was Apollo 13's mission?

7. Describe Apollo 13 spacecraft. Pay attention to shapes vocabulary.

8. What was the speed of the aircraft?

9. What was the problem?

10. What was the practical joke that Fred Hayes pulled on the crew?

11. What were the famous words used to report the problem?

LANGUAGE DEVICES: Concise Technical Writing. Phrasal Verbs. Verbs Become Nouns.

Phrasal verbs consist of a verb and a particle that together have a special meaning, for example, **break down** means “stop functioning properly”, **carry out** means “do”, **put off** means “postpone”.

In technical English phrasal nouns - nouns derived from phrasal verbs - are often used as one way to use the language concisely. When phrasal verbs become nouns, sometimes the verb comes first: **touchdown**, **check-up**, **cleanup** and sometimes the particle comes first: **overhaul**, **output**. If the verb comes first, one can either put a hyphen between the noun and the particle, e.g.: **build-up** or write it as one word, e.g.: **shutdown**. Hyphenated nouns might sometimes be considered more formal than one word.

READING I

Read the following text describing the problem onboard Apollo 13 and notice the phrasal verbs and nouns derived from phrasal verbs. Mark phrasal verbs in yellow and phrasal nouns in green.

At 5 1/2 minutes after liftoff, John Swigert, Fred Haise and James Lovell felt a little vibration. Then the center engine of the S-II stage shut down two minutes early. This caused the remaining four engines to burn 34 seconds longer than planned, and the S-IVB third stage had to burn nine seconds longer to put Apollo 13 in orbit.

Days before the mission, backup lunar module pilot, Charles Duke, inadvertently exposed the crew to German measles. Command Module Pilot Ken Mattingly had no immunity to measles and was replaced by backup command module pilot, John Swigert.

Ground tests before launch indicated the possibility of a poorly insulated supercritical helium tank in the lunar module, or LM, descent stage, so the flight plan was modified to enter the LM three hours early in order to obtain an onboard readout of helium tank pressure.

....During the first two days, the crew ran into a couple of minor surprises, but generally Apollo 13 was looking like the smoothest flight of the program. At 46 hours, 43 minutes Joe Kerwin, the capsule communicator, or Capcom, on duty, said, "The spacecraft is in real good shape as far as we are concerned. We're bored to tears down here." It was the last time anyone would mention boredom for a long time.

At 55 hours, 46 minutes, as the crew finished a 49-minute TV broadcast showing how comfortably they lived and worked in weightlessness, Lovell said, "This is the crew of Apollo 13 wishing everybody there a nice evening, and we're just about ready to close out our inspection of Aquarius and get back for a pleasant evening in Odyssey. Good night."

Nine minutes later, oxygen tank No. 2 blew up, causing the No. 1 tank to also fail. The command module's normal supply of electricity, light and water was lost, and they were about 200,000 miles from Earth.

...Thirteen minutes after the explosion, Lovell happened to look out of the left-hand window and saw the final evidence pointing toward potential catastrophe. "We are venting something out into the... into space," he reported to Houston. Capcom Jack Lousma replied, "Roger, we copy you venting." Lovell said, "It's a gas of some sort." It was oxygen gas escaping at a high rate from the second, and last, oxygen tank.

...At one hour, 29 seconds after the bang, Lousma said after instructions from Flight Director Glynn Lunney, "It is slowly going to zero, and we are starting to think about the LM lifeboat." Swigert replied, "That's what we have been thinking about too."

Ground controllers in Houston faced a formidable task. Completely new procedures had to be written and tested in the simulator before being passed up to the crew. The navigation problem had to be solved; essentially how, when and in what attitude to burn the LM descent engine to provide a quick return home.

...Power also was a concern. There were 2,181 ampere hours in the LM batteries. Ground controllers carefully worked out a procedure where the CM batteries were charged with LM power. All noncritical systems were turned off and energy consumption was reduced to 1/5, which resulted in having 20 percent of LM electrical power left when Aquarius was jettisoned. There was one electrical close call during the mission. One of the CM batteries vented with such force that it momentarily dropped off the line. Had the battery failed, there would have been insufficient power to return the ship to Earth.

Water was the main consumable concern. It was estimated that the crew would run out of water about five hours before Earth re-entry, which was calculated at around 151 hours. However, data from Apollo 11, which had not sent its LM ascent stage crashing into the moon as in subsequent missions, showed that its mechanisms could survive seven or eight hours in space without water cooling. The crew conserved water. They cut down to six ounces each per day, 1/5 of normal intake, and used fruit juices; they ate hot dogs and other wet-pack foods when they ate at all. The crew became dehydrated throughout the flight and set a record that stood up throughout Apollo: Lovell lost 14 pounds and the crew lost a total of 31.5 pounds, nearly 50 percent more than any other crew. Those stringent measures resulted in the crew finishing with 28.2 pounds of water, about 9 percent of the total.

Removal of carbon dioxide also was a concern. There were enough lithium hydroxide canisters, which remove carbon dioxide from the spacecraft, but the square canisters from the command module were not compatible with the round openings in the lunar module environmental system. There were four cartridges from the LM and four from the backpacks, counting backups. However, the LM was designed to support two men for two days and was being asked to care for three men for about four days. After a day and a half in the LM, a warning light showed that the carbon dioxide had built up to a dangerous level. Mission control devised a way to attach the CM canisters to the LM system by using plastic bags, cardboard and to tape all materials carried on board.

One of the big questions was, "How to get back safely to Earth?" The LM navigation system wasn't designed to help in this situation. Before the explosion at 30 hours, 40 minutes, Apollo 13 had made the normal midcourse correction, which would take it out of a free-return-to-Earth trajectory and put it on a lunar landing course. Now the task was to get back on a free-return course. The ground computed a 35-second burn and fired it five hours after the explosion. As they approached the moon, another burn was computed; this time a long five-minute burn to speed up the return home. It took place two hours after rounding the far side of the moon.

... The command module was cold and clammy at the start of power-up....

TOPIC-RELATED VOCABULARY

Rewrite the sentences using phrasal verbs or phrasal nouns from the text. Use the correct tense/form.

1. The oxygen tank *exploded* causing the oxygen to *escape*.

2. Apollo 13 *accelerated* due to a long burn computed by the ground.

3. There were dangerously high levels of the carbon dioxide *accumulating* in the LM.

4. The *amount of water consumption* had to be drastically *reduced*.

5. 5.5 minutes after *the rocket took off*, the crew felt some vibration.

READING II

Read the following text, look up the unknown words in the dictionary and be able to explain their meaning. Summarize the text in 10 sentences.

NASA, SpaceX Launch DART: First Test Mission to Defend Planet Earth

NASA's Double Asteroid Redirection Test (DART), the world's first full-scale mission to test technology for defending Earth against potential asteroid or comet hazards, launched Wednesday at 1:21 a.m. EST on a SpaceX Falcon 9 rocket from Space Launch Complex 4 East at Vandenberg Space Force Base in California.

Just one part of NASA's larger planetary defense strategy, DART – built and managed by the Johns Hopkins Applied Physics Laboratory (APL) in Laurel, Maryland – will impact a known asteroid that is not a threat to Earth. Its goal is to slightly change the asteroid's motion in a way that can be accurately measured using ground-based telescopes.

DART will show that a spacecraft can autonomously navigate to a target asteroid and intentionally collide with it – a method of deflection called kinetic impact. The test will provide important data to help better prepare for an asteroid that might pose an impact hazard to Earth, should one ever be discovered. LICIACube, a CubeSat riding with DART and provided by the Italian Space Agency (ASI), will be released prior to DART's impact to capture images of the impact and the resulting cloud of ejected matter. Roughly four years after DART's impact, ESA's (European Space Agency) Hera project will conduct detailed surveys of both asteroids, with particular focus on the crater left by DART's collision and a precise determination of Dimorphos' mass.

"DART is turning science fiction into science fact and is a testament to NASA's proactivity and innovation for the benefit of all," said NASA Administrator Bill Nelson. "In addition to all the ways NASA studies our universe and our home planet, we're also working to protect that

home, and this test will help prove out one viable way to protect our planet from a hazardous asteroid should one ever be discovered that is headed toward Earth.”

At 2:17 a.m., DART separated from the second stage of the rocket. Minutes later, mission operators received the first spacecraft telemetry data and started the process of orienting the spacecraft to a safe position for deploying its solar arrays. About two hours later, the spacecraft completed the successful unfurling of its two, 28-foot-long, roll-out solar arrays. They will power both the spacecraft and [NASA’s Evolutionary Xenon Thruster – Commercial ion engine](#), one of several technologies being tested on DART for future application on space missions.

“At its core, DART is a mission of preparedness, and it is also a mission of unity,” said Thomas Zurbuchen, associate administrator for the Science Mission Directorate at NASA Headquarters in Washington. “This international collaboration involves DART, ASI’s LICIAcube, and ESA’s Hera investigations and science teams, which will follow up on this groundbreaking space mission.”

DART’s one-way trip is to the Didymos asteroid system, which comprises a pair of asteroids. DART’s target is the moonlet, Dimorphos, which is approximately 530 feet (160 meters) in diameter. The moonlet orbits Didymos, which is approximately 2,560 feet (780 meters) in diameter.

Since Dimorphos orbits Didymos at much a slower relative speed than the pair orbits the Sun, the result of DART’s kinetic impact within the binary system can be measured much more easily than a change in the orbit of a single asteroid around the Sun.

“We have not yet found any significant asteroid impact threat to Earth, but we continue to search for that sizable population we know is still to be found. Our goal is to find any possible impact, years to decades in advance, so it can be deflected with a capability like DART that is possible with the technology we currently have,” said Lindley Johnson, planetary defense officer at NASA Headquarters. “DART is one aspect of NASA’s work to prepare Earth should we ever be faced with an asteroid hazard. In tandem with this test, we are preparing the Near-Earth Object Surveyor Mission, a space-based infrared telescope scheduled for launch later this decade and designed to expedite our ability to discover and characterize the potentially hazardous asteroids and comets that come within 30 million miles of Earth’s orbit.”

The spacecraft will intercept the Didymos system between Sept. 26 and Oct. 1, 2022, intentionally slamming into Dimorphos at roughly 4 miles per second (6 kilometers per second). Scientists estimate the kinetic impact will shorten Dimorphos’ orbit around Didymos by several minutes. Researchers will precisely measure that change using telescopes on Earth. Their results will validate and improve scientific computer models critical to predicting the effectiveness of the kinetic impact as a reliable method for asteroid deflection.

“It is an indescribable feeling to see something you’ve been involved with since the ‘words on paper’ stage become real and launched into space,” said Andy Cheng, one of the DART

investigation leads at Johns Hopkins APL and the individual who came up with the idea of DART. “This is just the end of the first act, and the DART investigation and engineering teams have much work to do over the next year preparing for the main event – DART’s kinetic impact on Dimorphos. But tonight we celebrate!”

DART’s single instrument, the Didymos Reconnaissance and Asteroid Camera for Optical navigation (DRACO), will turn on a week from now and provide first images from the spacecraft. DART will continue to travel just outside of Earth’s orbit around the Sun for the next 10 months until Didymos and Dimorphos will be a relatively close 6.8 million miles (11 million kilometers) from Earth.

A sophisticated guidance, navigation, and control system, working together with algorithms called Small-body Maneuvering Autonomous Real Time Navigation (SMART Nav), will enable the DART spacecraft to identify and distinguish between the two asteroids. The system will then direct the spacecraft toward Dimorphos. This process will all occur within roughly an hour of impact.

Johns Hopkins APL manages the DART mission for NASA's Planetary Defense Coordination Office as a project of the agency’s Planetary Missions Program Office. NASA provides support for the mission from several centers, including the Jet Propulsion Laboratory in Southern California, Goddard Space Flight Center in Greenbelt, Maryland, Johnson Space Center in Houston, Glenn Research Center in Cleveland, and Langley Research Center in Hampton, Virginia. The launch is managed by NASA’s Launch Services Program, based at the agency’s Kennedy Space Center in Florida. SpaceX is the launch services provider for the DART mission.

Source: <https://www.nasa.gov/press-release/nasa-spacex-launch-dart-first-test-mission-to-defend-planet-earth> retrieved: 2021-12-13

Read the statements and decide whether they are true (T), false (F) or not given (NG)

1. NASA’s Double Asteroid Redirection Test (DART) purpose is to collide asteroids that might pose an impact hazard to Earth.
2. After DART's impact, ESA's (European Space Agency) Hera project will analyze the crater left by DART's collision and an accurate determination of the mass of Didymos.
3. NASA’s Double Asteroid Redirection Test will help prove out one viable way to protect our planet from a hazardous asteroid should one ever be discovered that is headed toward Earth.
4. Following the separation from the second stage of the rocket, after just 10 minutes, mission operators received the first spacecraft telemetry data and started the process of orienting the spacecraft to a safe position for deploying its solar arrays.
5. According to Thomas Zurbuchen, the international collaboration involves DART, ASI’s LICIACube, ESA’s Hera investigations and science teams.

6. DART's target is the moonlet, Dimorphos, which is approximately 530 feet (160 meters) in diameter.
7. According to Lindley Johnson, several asteroids posing a significant threat to Earth have been already found, and the goal is to find more possible impacts, years to decades in advance.
8. The spacecraft will intercept the Dimorphos system between Sept. 26 and Oct. 1, 2022.
9. DART will travel around the Sun for the next 10 months until Didymos and Dimorphos will be at approximately 11 million kilometers from Earth.
10. DART spacecraft will distinguish between Didymos and Dimorphos with the help of a sophisticated guidance, navigation, and control system, working together SMART Nav.

READING III

THE PRIVATE SPACE RACE

A. On 30 May 2020, two astronauts - Doug Hurley and Bob Behnken - lifted off on their way to the International Space Station. After a stay of two months, on 2 August, the so-called Demo-2 mission splashed safely into The Gulf of Mexico. Not exactly big news, you may be forgiven for thinking. After all, the ISS has been circling the Earth for over 20 years now during which time it has received some 240 visitors. Bob and Doug's trip, however, wasn't about the destination; it was the way they got there that made it so special.

First of all, the two were riding a dragon: The Crew Dragon capsule is the human-rated update to SpaceX's highly successful Cargo Dragon spacecraft, which supplies the ISS on a regular basis. The Crew Dragon also happens to be the first new American spacecraft carrying astronauts into orbit in 39 years.

Its direct predecessor was the famous Space Shuttle, which served as a space workhorse from 1981 until its retirement in 2011. Since then, NASA has had to grudgingly buy seats in Russian Soyuz capsules to get their astronauts to the ISS. Another reason, at least for Americans, to celebrate the smooth Demo-2 mission: it marked the return to flight from American soil and the end of a long wait - especially for Doug Hurley, who was also the pilot of the final Shuttle mission.

Hurley's old and new ride couldn't be more different. The Shuttle was a truck, the Crew Dragon is a sports car. Instead of hundreds of knobs, switches and gauges the Dragon has only a handful of physical controls and is operated mainly via three large touchscreen panels. Where the Shuttle had fighter-jet-like seats the Dragon welcomes astronauts into four slim bucket seats surrounded by smooth white panels instead of science racks. Bob and Doug's pressure suits look like fashionable white motorcycle gear and not something that's called 'pumpkin suit' for nothing. At first sight, the SpaceX Crew Dragon may feel a bit like one of those show cars at a car fair: its cool and sleek look lets you dream of the future but you know exactly that nothing of all this will carry over to the drab production model. Not this one, the Crew Dragon is the production model ... and the future; the future not just for NASA, which finally has a spacecraft again to

carry astronauts into orbit. While technically very impressive and, at least for some of us, of patriotic importance, this flight was also the first time a privately owned company had carried humans into space. Bob and Doug may have been official NASA astronauts but they were flying in a spacecraft that was procured from SpaceX, which had independently designed, engineered and built the capsule as well as the Falcon 9 rocket lifting it into space.

B. Before, NASA had done the main design in-house and contracted the manufacturing without a proper cost cap. A very costly process which, although it had put men on the Moon, was getting more and more inefficient. To introduce a bit of healthy free-market competition and reduce the costs for supplying the ISS NASA started the Commercial Orbital Transportation Services program in 2006. COTS awarded fixed funds to private companies, which reached certain predefined milestones on the road to space. The two finalists, SpaceX and Orbital Science Corporation were then contracted to fly cargo to the ISS on a regular basis.

The program was so effective that, in 2010, a similar process, the Commercial Crew Development program or CCDev, was initiated to restore crew transport to the ISS after the retirement of the Space Shuttle. Now, of course, the stakes – and the awards - were much higher. Who would be the first private company to fly humans into space?

The initial competitors were Sierra Nevada Corporation with the Dream Chaser, a winged spacecraft looking like a mini version of the Shuttle, SpaceX with the Crew Dragon and Boeing with the CST-100 Starliner, a classic cone-shaped design that, although reminiscent of an Apollo command module, is capable of landing on land.

Sierra Nevada dropped out in 2017 after NASA lost faith in a winged spacecraft and cut the funding. Now it was 'old space' vs. 'new space'. Boeing, of course, had built the first stage of the Saturn V moon rocket, the lunar rovers, the American modules for the ISS and provided launch services for the Space Shuttle - a multi-billion corporation with huge resources and experience. SpaceX, on the other hand, had only been in space since 2008 when its Falcon 1 reached orbit on the fourth attempt - consequently saving the company from bankruptcy. In more recent years, however, SpaceX has become rather successful in providing affordable launch services for satellites. Their Falcon Heavy is also currently the most powerful rocket.

Despite these discrepancies, at the beginning of 2018, both companies were neck and neck with important milestones scheduled only weeks apart. Then Boeings deadlines started slipping and, in March 2019, SpaceX took the lead with a picture perfect demo mission that automatically docked an uncrewed Crew Dragon at the ISS. The success, however, was short lived: a month later the capsule was completely destroyed by an explosion during a static fire test. Investigations into the cause would set back SpaceX for months if not years.

This was Boeing's chance. In December 2019, they launched their own uncrewed orbital flight test to the ISS or at least tried to. A timer malfunction prevented the Starliner from reaching the correct orbit and the mission had to be aborted. Another software bug which would have caused a crash between the service and command module was luckily detected shortly before their separation. Although the Starliner touched down safely in the desert of New Mexico, the flight was deemed a failure.

A month later, SpaceX had a surprise comeback. In record time they had found the cause for the capsule's explosion. To celebrate, they launched a very special kind of firework. The

inflight abort test, a critical milestone, would demonstrate the private spaceship's ability to separate from a malfunctioning and eventually exploding booster shortly after launch. While Boeing was allowed to demonstrate their inflight abort in a simulation with a digital twin, SpaceX went for the real thing.

A nail-biting experience, especially for the engineers since the thrusters which would pull the Crew Dragon away from the Falcon 9 booster had also caused the explosion nine months earlier. What a relief when the mighty Dragon safely pulled away before the booster erupted in a giant plume. Finally, the road was clear for Bob and Doug's flight.

Their Demo-2 mission might seem like the end of the private space race but it is only just the beginning for a new age in space travel. Boeing will repeat their bodged uncrewed flight and sooner rather than later fly NASA astronauts to the ISS as well. And not just NASA astronauts. Once the paperwork is cleared Crew Dragon and Starliner can also be piloted by employees of their respective companies and ferry civilian customers into orbit and to the ISS. Private 'space cruise' operator

Axion Space is already selling tickets and planning a private space hotel module for the ISS.

While the rumoured 55 million USD per ticket might not exactly be within everybody's reach, a second private space race on a smaller scale is promising astronaut wings for as low as 250,000 USD - basically a steal.

C. The beginning of space is defined by the somewhat arbitrary Kármán line at an altitude of 100 kilometres. Everyone crossing this line can call themselves an astronaut - even if they drop straight down again after reaching the magic mark. Naturally, these suborbital hops are much easier to accomplish than the 27,000 kph necessary for reaching orbit.

Currently, Virgin Galactic and Blue Origin are competing to bring space tourism to the upper ten - and not just the upper one. Virgin Galactic seemed to have a strong head start. Their concept is based on Spaceship One, which in 2004 won the X Prize for the first crewed private space flight. The consequently dubbed Spaceship Two is a small eight-seater spaceplane which is carried to its launch altitude by a twin-fuselage, jet-powered aircraft, the White Knight Two. Once dropped it ignites its rocket engine to reach the edge of space. After a couple of minutes weightlessness and some mind-blowing views at apogee it folds its wings for re-entry and lands on a normal runway.

Development of the unusual but elegant concept, however, was almost brought to a standstill in 2014 when one of the test pilots was killed after an in-flight breakup of the spaceplane.

This was Blue Origin's chance to catch up. Since 2011 they have been conducting tests with their New Shepard rocket - named after Alan Shepard, the first American in space. New Shepard is a more classical approach: the rocket consists of a small, stubby single-engine booster and crew capsule with large windows and room for six passengers. After a vertical lift-off, separation and some precious minutes of 'space time' the capsule deploys its parachutes and lands in the desert - as demonstrated in November 2015 with a successful test flight.

After Space Ship Two's return to flight, Virgin Galactic and Blue Origin are currently neck and neck again and the first passenger flights are expected in the next couple of months. The private space race for suborbital flights is still open.

D. A win for Blue Origin would be their second. They won a major stage in another much more lucrative space race – the race for reusability. After deploying the capsule in that 2015 flight, the New Shepard booster did something which had never been done before: after a little drop it relit its engine, unfolded landing legs, and softly touched down next to the launch pad, ready for reuse. Blue Origin had beaten SpaceX, who had been trying to land one of their Falcon 9 booster's in one piece for a while now, by a month.

Then again, the comparison is a bit misleading: there's quite a difference between catching the fall of a little rocket or yanking a 48-meter-tall booster that's travelling at a speed of some 7000 kph back to the launch complex - or a barge floating in the ocean for that matter. Since their little defeat, SpaceX has perfected the art of landing a booster, be it on land, on sea - or three at a time in the case of a Falcon Heavy with its three identical boosters. And not just landing them but also reusing them up to six times, which drops launch costs dramatically. After some 100 launches and 60 landings of a Falcon 9 booster, SpaceX is currently one of the most competitive launch providers conducting 20 percent of all launches worldwide.

A profitable business that Blue Origin wants a share of, too. At the moment, they are working on their next rocket, the New Glenn. With a reusable first stage the rocket will be capable of lifting 45 tonnes into low Earth orbit - about 1.5 times the capability of the current record holder, SpaceX's Falcon Heavy. First test flights are scheduled for 2021 but Blue Origin is working in secrecy and so far no flight hardware has been spotted except for a few of their new rocket engines.

Certainly no time for SpaceX to rest on its laurels. Literally in a field in Texas and for everybody to see, they are already working on an even bigger rocket, the Starship. It will again double the lift capability and not only the booster but also the upper stage will be reusable. On top of each other they will even top the mighty Saturn V moon rocket in size.

The private space race has become much more than just the race for the money of a few super-rich space tourists. The next generation of reusable rockets will dramatically lower the costs to reach space - for everybody. Who knows, one day a ticket to space might cost no more than a long-distance flight in first class. But that's another story for the future. The future of private space.

Matthias Maier. INCH, Nr. 26, 2020. Pp. 30-35.

Task 1. Choose the correct heading for each paragraph (A-D) from the list below.

- *Space jumps big and small*
- *Reduce, reuse, recycle*
- *Space Shuttle's successor in action*
- *Rocking the launch business*

Paragraph A

Paragraph B

Paragraph C

Paragraph D

Task 2. Find words in the text with the same or similar meanings as the following.

Paragraph A

1. To leave the ground and rise into the air (especially about a spaceship):
2. Unwillingly, reluctantly:
3. Dull, boring to look at or experience:

Paragraph B

1. To be level with each other and have an equal chance of winning:
2. To stop a process, plan, or activity before it has been completed:
3. A large quantity of smoke that rises into the air in a column:

Paragraph C

1. The highest or greatest point:
2. Shorter and thicker than usual:
3. An extra engine in a machine such a space rocket, which provides an extra amount of power at certain times:

Paragraph D

1. Profitable:
2. Pull suddenly and with a lot of force:
3. To be satisfied with one's achievements and not to make an effort to do anything else:

If you would like to find out more about SpaceX's successful landings, watch the following video: <https://www.youtube.com/watch?v=Wn5HxXKQOjw>
(How SpaceX Lands Rockets with Astonishing Accuracy, 2019. Art of Engineering. 11 minutes).

If you would like to enjoy the sight of Falcon Heavy launch and landing, mentioned in the article, click here: <https://www.youtube.com/watch?v=mY-fSnKTLqw>

Instructions

The purpose of instructions, also called procedural texts, is to tell the reader how to do or make something. The information is presented in a logical sequence of steps required to complete a task. These texts are usually written in **the present simple tense** or **in imperatives**. The most common every-day example of a procedural text is a recipe. There are different types of procedural texts for different purposes:

- Texts that explain how to use something, i.e. instruction/operation manuals, e.g., how to use a computer or a photocopier.

Instructions need to be clear, logical and easy to follow:

- The goal should be clearly stated.
- The materials should be listed in order of use.
- The layout of the text should be easy to understand and follow (steps should be numbered or listed).
- The steps should be in chronological order.

LANGUAGE DEVICES FOR INSTRUCTIONAL WRITING

The use of the imperative. The imperative has the same form as the infinitive form of the verb.

Open the book.
Insert the tube.
Calculate the result of the equation.

The use of the passive form.

A container *is filled* with a mixture of salt and water.
The defective part should be *replaced*, and then the machine should *be reconnected*.

The use of modal verbs.

The level of lubricant *must* be checked after every 100 hours of operation.
The valve *should* be left in the open position.
The bearings *have to* be inserted after the housing has been cleaned.

Language to talk about simultaneous actions.

Simultaneously, at the same time, while doing something, as you are doing something...

Exercise 1 Complete the sentences below with verbs commonly used in operating instructions.

1. _____ the timer to 5 minutes.
2. _____ the temperature of the oven.
3. _____ the button, _____ it down and _____ it.
4. _____ the coin and _____ the lever.
5. _____ the knob three times.
6. _____ the temperature on the gauge.
7. _____ the key from the lock and _____ it on the table.
8. _____ "Open" from the "File" menu.

9. _____ the instructions on the back of the machine.
 _____ on the icon, _____ it to the trash can,
 and _____ it into the trash can.

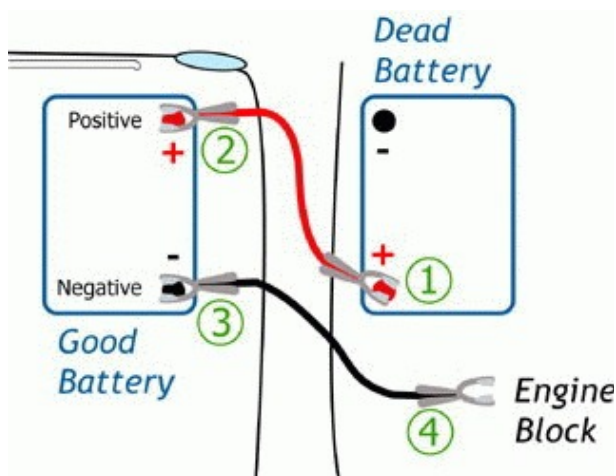
Exercise 2: Complete the following chart.

NOUN	ADJECTIVE	VERB
	loose	to loosen
position		
load		
	stiff	
seal		
	short	
	tight	
	weak	
clamp		
strength		
lever		
screw		
grease		

Exercise 3

A Dead Battery

It's another cold morning in Karlsruhe and your battery is dead. Write a description of how to jumpstart a car using a passing motorist's battery--without ruining either your car or the other motorist's car! Use **sequencing terms** (e.g., first, then, thereafter, subsequently, finally).



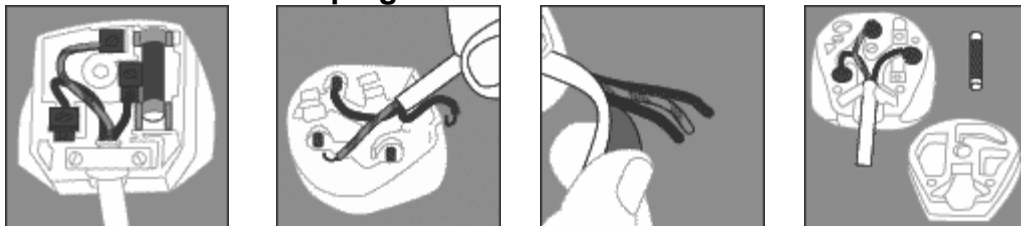
a battery terminal
 the plus terminal / the positive terminal
 the minus terminal / the negative terminal
 a dead battery ↔ a charged battery
 to connect a cable to ground
 the rated voltage of a battery (*Nennspannung*)
 jumper cables
 the red cable
 the black cable
 to jumpstart a car

to connect ↔ to disconnect
 to flow (electric current flows)
 to start the engine
 to keep the engine running

Source: http://www.t4y.info/wp-content/uploads/2011/01/How_To_Jump_Start_A_Car.jpg

Exercise 4

How to wire a British plug



Step 1: Open up the plug

Use a medium-sized screwdriver to undo the screw in the center of the underside of the plug.

Remove the plug's top cover but take care none of the pins or the fuse fall out. Some plugs also have two smaller screws that hold a clamp across the wire. This stops the cord being pulled out of the plug. Loosen these screws and then loosen the brass screws that hold each wire into the pin. There will be either two or three wires to loosen.

Step 2: Prepare the wires

Place the wire inside the plug and mark how long it needs to be to reach the terminals. Score the outer sheath being careful not to cut the wires on the inside.

Modern wiring is colored: blue for neutral, brown for live and green/yellow for earth. The new plug will have clear markings on the plastic case for E (earth), L (live) and N (neutral).

Step 3: Cut away the tips

Cut the wires to the length required and use wire strippers or pliers to pull away 12mm (1/2in) of insulation from the end of each wire.

If the wires are frayed or damaged in any way, cut through the cord and strip away enough sheathing to make new wire connections.

Twist the ends of each wire (this will make for a better connection) and push into the terminals. Then use a small screwdriver to secure each wire in place. Some plugs have a cylindrical post terminal with a hole for the wire. All the wires must be inside the hole and firmly held by the top screw.

Double check you have the correct color of wire connected to each terminal.

Tip: Always make sure the wires lie flat and aren't trapped across part of the plug casing.

Step 4: Fit the casing

The cord is held in place by a clamp and screws or a plastic grip. Fit the cord into the plug so that all the sheathed flex is tightly held in place.

Check you have the correct fuse for the appliance fitted and screw the top of the plug back on. Test the appliance.

Exercise 5

- a) **Work individually.** Choose a short procedure that you know something about and write a set of instructions to explain how to do it. Where appropriate include simultaneous actions.

b) **Work in pairs.** Take turns to be the instructor and the trainee. Use phrasal verbs and language for instructions wherever possible. Being a trainee, respond and ask questions as appropriate.



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